

Presentation at the World Congress of the Econometric Society: Extreme Points and Majorization

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1 Introduction

This presentation provides an overview of how majorization and extreme point characterizations can be applied to solve various economic design problems. We begin by reviewing the concept of majorization and its role in economics, then present the main characterization results from Kleiner, Moldovanu, and Strack (2021) for extreme points of sets defined by monotonicity and majorization constraints. The key insight is that many seemingly different economic problems share a common mathematical structure: they involve maximizing a functional over sets of monotonic functions that are either majorized by, or majorize, a given function. We demonstrate how our characterization results can be applied to auction design, contest design, information design, optimal delegation and optimal stopping problems.

For each application, we explain why the problem falls within our framework and how extreme point characterizations are a useful analytical tool. We conclude by surveying recent related work that extends these characterizations to settings with additional constraints, multidimensional spaces, and alternative orderings.

2 Extreme Points and Majorization

2.1 Majorization Preliminaries

Throughout, we let L^1 denote the real-valued and integrable functions defined on $[0, 1]$ endowed with the L^1 -norm.¹

¹Formally, $f \in L^1$ is an equivalence class of functions that are equal almost everywhere. An equivalence class $f \in L^1$ is nondecreasing if it contains a nondecreasing element. Whenever f is nondecreasing, it has a nondecreasing and right-continuous element, which we use as a canonical representative.

Motivated by economic applications, we restrict attention throughout to nondecreasing functions. This monotonicity constraint arises for various reasons in applications: for example, since cumulative distribution functions are nondecreasing, the monotonicity constraint arises in information design problems; since incentive-compatible mechanisms often correspond to nondecreasing allocations, the monotonicity constraint arises in various mechanism design problems.

For two nondecreasing functions $f, g \in L^1$ we say that f *majorizes* g , denoted by $f \succ g$ if the following two conditions hold:

$$\begin{aligned} \int_x^1 g(s) \, ds &\leq \int_x^1 f(s) \, ds \quad \text{for all } x \in [0, 1] \text{ and} \\ \int_0^1 g(s) \, ds &= \int_0^1 f(s) \, ds. \end{aligned} \tag{1}$$

We say that f *weakly majorizes* g , denoted by $g \prec_w f$, if the first condition above holds (but not necessarily the second).²

Majorization constraints arise for various reasons in economic applications. We will see how this constraint captures resource constraints on the available objects and prizes in auction and contest design, constraints on feasible information structures in information design, and constraints induced by the absence of transfers in delegation problems.

This motivates the study of the subset of nondecreasing functions that are majorized by, or majorize, a given function:

$$\text{MPS}(f) = \{g \in L^1 \mid g \text{ nondecreasing such that } f \succ g\}.$$

Similarly, we denote by $\text{MPS}_w(f)$ the set of nonnegative, nondecreasing functions that are weakly majorized by f . Finally, let

$$\text{MPC}(f) = \{g \in L^1 \mid g \text{ nondecreasing such that } g \succ f \text{ and } f(0) \leq g \leq f(1)\}.$$

An *extreme* point of a convex subset A is a point $x \in A$ that cannot be represented as a convex combination of two distinct points in A . Formally, if X is a vector space and $A \subseteq X$ is convex, $x \in A$ is an extreme point of A if $x = \alpha y + (1 - \alpha)z$, for $z, y \in A$ and $\alpha \in [0, 1]$ imply together that $y = x$ or $z = x$.

In the following, we characterize the extreme points of $\text{MPS}(f)$, $\text{MPS}_w(f)$, and $\text{MPC}(f)$. It turns out that the characterization of these extreme points is particularly tractable and

²We will only consider the majorization relation for nondecreasing functions. For nonmonotone functions f and g , in the literature, f is defined to majorize g if the nondecreasing rearrangement of f majorizes the nondecreasing rearrangement of g .

can be used in at least three ways in various economic applications:

1. **Information about extreme points directly gives qualitative insights into solutions of optimization problems:** *Bauer's Maximum Principle* says that a convex, upper-semicontinuous functional on a non-empty, compact and convex set A of a locally convex space attains its maximum at an extreme point of A . Therefore, for many optimization problems a solution can be found at an extreme point, and properties common to all extreme points are therefore satisfied by an optimal solution.
2. **Characterization of extreme points is useful to derive results characterizing when particular extreme points solve a linear optimization problem:** Propositions 1 and 2 characterize when a given extreme point solves a linear optimization problem with constraint set $\text{MPS}(f)$ and $\text{MPC}(f)$, respectively.
3. **Many properties of extreme points are inherited by arbitrary elements of the underlying set:** The *Krein–Milman Theorem* states that any convex and compact set A in a locally convex space is the closed, convex hull of its extreme points. Therefore, any element of such a set is a limit of convex combinations of extreme points. Alternatively, *Choquet's Theorem* gives conditions such that any element is a mixture of extreme points.³ Therefore, if all extreme points have a common property and this property is preserved by taking convex combinations and limits (or taking mixtures), then every element of the set has this property. We repeatedly use this insight below; for example, in an auction application we show that every extreme point of $\text{MPS}_w(f)$ is the interim allocation rule of a BIC auction and deduce from this that the set $\text{MPS}_w(f)$ characterizes the set of interim allocation rules of BIC auctions. For the purpose of the presentation, we will pretend that all mixtures are actually convex combinations.

The following result characterizes the extreme points of $\text{MPS}(f)$:

Theorem 1. *Let f be nondecreasing. Then g is an extreme point of $\text{MPS}(f)$ if and only if there exists a collection of disjoint intervals $[\underline{x}_i, \bar{x}_i)$ indexed by $i \in I$ such that for a.e. $x \in [0, 1]$*

$$g(x) = \begin{cases} f(x) & \text{if } x \notin \bigcup_{i \in I} [\underline{x}_i, \bar{x}_i) \\ \frac{\int_{\underline{x}_i}^{\bar{x}_i} f(s) \, ds}{\bar{x}_i - \underline{x}_i} & \text{if } x \in [\underline{x}_i, \bar{x}_i). \end{cases} \quad (2)$$

Intuitively, if a function g is an extreme point of $\text{MPS}(f)$ then, at any point in its domain, either the majorization constraint binds, or the monotonicity constraint binds. This implies either that $g(x) = f(x)$ or that g is constant at x .

³We say that an element x of a metric space is a mixture of a set A if there a probability distribution with support A whose barycenter is x .

The previous result can be used to characterize the extreme points of $\text{MPS}_w(f)$. For $A \subseteq [0, 1]$, denote by $\mathbf{1}_A(x)$ the indicator function of A : it equals 1 if $x \in A$ and it equals 0 otherwise.

Corollary 1. *Suppose that f is nondecreasing and nonnegative. A function g is an extreme point of $\text{MPS}_w(f)$ if and only if there is $\theta \in [0, 1]$ such that g is an extreme point of $\text{MPS}(f \cdot \mathbf{1}_{[\theta, 1]})$ and $g(x) = 0$ for a.e. $x \in [0, \theta]$.*

Finally, we characterize the extreme points of $\text{MPC}(f)$.

Theorem 2. *Let f be nondecreasing and continuous. Then $g \in \text{MPC}(f)$ is an extreme point of $\text{MPC}(f)$ if and only if there exists a collection of intervals $[\underline{x}_i, \bar{x}_i)$, (potentially empty) sub-intervals $[\underline{y}_i, \bar{y}_i) \subset [\underline{x}_i, \bar{x}_i)$, and numbers v_i indexed by $i \in I$ such that for a.e. $x \in [0, 1]$*

$$g(x) = \begin{cases} f(x) & \text{if } x \notin \bigcup_{i \in I} [\underline{x}_i, \bar{x}_i) \\ f(\underline{x}_i) & \text{if } x \in [\underline{x}_i, \underline{y}_i) \\ v_i & \text{if } x \in [\underline{y}_i, \bar{y}_i) \\ f(\bar{x}_i) & \text{if } x \in [\bar{y}_i, \bar{x}_i) \end{cases} \quad (3)$$

Extreme Points: An Intuitive Description Let f be a cumulative distribution function (CDF) and recall that a CDF admits a jump at a given value if the distribution assigns a mass point to that value. As h majorizes g if and only if g is a mean-preserving spread of h , it follows that $\text{MPS}(f)$ is the set of *mean preserving spreads* of f and $\text{MPC}(f)$ is the set of *mean preserving contractions* of f . These properties are also reflected in the extreme points: Each extreme point $g \in \text{MPS}(f)$ is obtained by taking the mass in each interval $[\underline{x}_i, \bar{x}_i]$ and spreading it out into two mass points at the boundaries of the interval, $\underline{x}_i$ and $\bar{x}_i$. There is a unique way to do so while preserving the mean determined by (2). In contrast, each extreme point $g \in \text{MPC}(f)$ is obtained by contracting the mass in each interval $[\underline{x}_i, \bar{x}_i]$ into two mass points placed at $\underline{y}_i$ and $\bar{y}_i$.

The main insight of Theorems 1 and 2 is that the mean-preserving spreads (or contractions) of f described there cannot be represented as convex combinations of other functions in MPS and in MPC, respectively, and that these are the only functions with this property.

3 Linear Optimization under Majorization Constraints

We now consider optimization problems where the objective is a linear functional, and where the constraint set is defined by majorization and monotonicity constraints.

3.1 Maximizing a Linear Functional on $\text{MPS}(f)$

Given a nondecreasing function f and a bounded function c consider the problem

$$\max_{h \in \text{MPS}(f)} \int_0^1 c(x)h(x) \, dx. \quad (4)$$

There are three cases:

1. If c is nondecreasing, an optimal g will be as large as possible for high values of x . From the majorization constraint it follows that f itself is a solution to the optimization problem.
2. If c is nonincreasing, an optimal g will be as large as possible for low values of x . From the monotonicity constraint it follows that the constant function $g(x) = \int_0^1 f(s) \, ds$ for all x is a solution.
3. If c is not monotonic, other extreme points of $\text{MPS}(f)$ may be optimal.

The next result essentially characterizes the conditions under which an arbitrary extreme point is optimal. As explained below, the ironing technique, originally used in Myerson (1981) for an optimization problem formulated without majorization constraints, can be used if the constraint set is MPS .

Define

$$C(x) = \int_0^x c(s) \, ds$$

and let $\text{co } C$ denote the convex hull of C , i.e., the largest convex function that lies below C .

Proposition 1. *Let g be an extreme point of $\text{MPS}(f)$, and let $\{[\underline{x}_i, \bar{x}_i) | i \in I\}$ be the collection of intervals described in Theorem 1. If $\text{co } C$ is affine on $[\underline{x}_i, \bar{x}_i)$ for each $i \in I$ and if $\text{co } C = C$ otherwise, then g is optimal. Moreover, if f is strictly increasing then the converse holds.*

3.2 Maximizing a Linear Functional on $\text{MPC}(f)$

We now analyze the problem

$$\max_{h \in \text{MPC}(f)} \int_0^1 c(x)h(x) \, dx. \quad (5)$$

Again, there are three cases:

1. If c is nonincreasing, an optimal solution is as large as possible for small values of x . It follows from the majorization constraint that f solves this problem.
2. If c is nondecreasing, an optimal solution is as large as possible for high values of x . It follows from the majorization constraint and the boundary constraint that an optimum

is obtained at the step function g defined by

$$g(x) = \begin{cases} f(0) & \text{for } x < \bar{x} \\ f(1) & \text{for } x \geq \bar{x}, \end{cases}$$

where $\bar{x}$ solves

$$\int_0^{\bar{x}} f(0) \, ds + \int_{\bar{x}}^1 f(1) \, ds = \int_0^1 f(s) \, ds$$

3. If c is nonmonotonic we can use our previous characterization together with the duality results in Dworczak and Martini (2019) to determine when particular extreme points are optimal.

Proposition 2. *Suppose that f is strictly increasing, and that f and c are continuous. Let g be an extreme point of $\text{MPC}(f)$, and let $\{\underline{x}_i, \bar{x}_i\}_{i=1}^n$ and $\{\underline{y}_i, \bar{y}_i\}_{i=1}^n$ be finite collections of intervals as described in Theorem 2 that satisfy $\bar{x}_i < \underline{x}_{i+1}$.*

Then g is optimal if and only if

1. *the complement of $\bigcup_{i \in I} [\underline{x}_i, \bar{x}_i]$ is a subset of the set where c is nonincreasing,*
2. *$c(\underline{y}_i)(x - \underline{y}_i) \leq \int_{\underline{y}_i}^x c(t) \, dt$ for all $i \in I$ and $x \in [\underline{x}_i, \bar{x}_i]$, and*
3. *equality holds in the previous inequality whenever $x = \underline{x}_i, \bar{y}_i, \bar{x}_i$ such that $x \neq 0, 1$.*

3.3 How the two problems differ

While Propositions 1 and 2 both give essentially a characterization for when particular extreme points are optimal, the two problems (4) and (5) differ fundamentally and distinct tools are needed to solve these problems: for problem (4) with constraint set MPS, we can use an ironing procedure to find an explicit solution, whereas for problem (5) with constraint set MPC we need to use a guess-and-verify approach. To illustrate these differences, we contrast the two problems:

$$\max_{g \in \text{MPS}(f)} \int_0^1 c(x)g(x) \, dx \qquad \max_{g \in \text{MPC}(f)} \int_0^1 c(x)g(x) \, dx$$

Note that we can impose without loss of generality that $g(1) = f(1)$ for any solution. Defining $C(x) = -\int_0^x c(s) \, ds$ and using integration by parts, and eliminating the terms $C(1)f(1)$ (which are constant in g), these problems can be written as:

$$\max_{g \in \text{MPS}(f)} \int_0^1 C(x) \, dg(x) \qquad \max_{g \in \text{MPC}(f)} \int_0^1 C(x) \, dg(x)$$

The corresponding dual linear programs are given by (see, for example, Shapiro, 2010; Dworczak and Martini, 2019):

$$\begin{array}{ll} \min_{p \text{ concave}} \int_0^1 p(x) \, df(x) & \min_{p \text{ convex}} \int_0^1 p(x) \, df(x) \\ \text{s.t. } p \geq C & \text{s.t. } p \geq C \end{array}$$

Moreover, strong duality holds, and therefore g is an optimal solution to the primal problem if and only if there exists a feasible solution to the corresponding dual problem that achieves the same value.

Note that the dual problem on the left is easy to solve: We aim to find a concave function that lies above a given function C and minimizes the integral. The concave envelope $\hat{C}$ of C is the pointwise smallest concave function that lies above C and therefore solves the dual problem. It follows that $g \in \text{MPS}(f)$ solves the primal problem if and only if both problems attain the same value: $\int C(s) \, dg(s) = \int \hat{C}(s) \, df(s)$. This implies that g , when interpreted as a probability measure, puts zero mass on points where $\hat{C}(s) < C(s)$. Letting $\{(\underline{x}_i, \bar{x}_i) : i \in I\}$ denote a maximal collection of maximal intervals on which $\hat{C} < C$ and observing that $\hat{C}$ is affine when restricted to $[\underline{x}_i, \bar{x}_i]$, it follows that a solution is given by the extreme point

$$g(x) = \begin{cases} f(x) & \text{if } x \notin \bigcup_{i \in I} [\underline{x}_i, \bar{x}_i] \\ \frac{\int_{\underline{x}_i}^{\bar{x}_i} f(s) \, ds}{\bar{x}_i - \underline{x}_i} & \text{if } x \in [\underline{x}_i, \bar{x}_i]. \end{cases}$$

Hence, an ironing technique analogous to Myerson provides a constructive method to obtain a solution.

Contrast this situation with the optimization problem with constraint set MPC: The dual problem is given on the right and is not as immediate to solve. In general, there is no pointwise smallest convex function above C if C is not convex. This explains why no analogous procedure to ironing is known to solve problems with the constraint set MPC. Linear programming duality can still be used to verify whether a particular extreme point solves a given problem, as illustrated in Proposition 2. However, it does not provide a direct method to construct a solution to a specific problem.

4 Applications

4.1 Allocation problems

We illustrate the usefulness of the extreme-point characterization obtained in the previous section by showing that it immediately implies a generalization of the symmetric version of Border's (1991) theorem and the BIC-DIC equivalence for symmetric mechanisms (Manelli and Vincent (2010), Gershkov, Goeree, et al. (2013)).

There are n agents with types $\theta_1, \dots, \theta_n$ that are independently and identically distributed on $[0, 1]$ according to a common distribution F , with density $f > 0$. Each agent wants at most one object. There are n objects with qualities $0 \leq q_1 \leq q_2 \leq \dots \leq q_n = 1$, and we define $\mathcal{A} \subset \{0, q_1, q_2, \dots, q_n\}^n$ to be the set of feasible allocations, i.e. $\alpha_i = q_k \neq 0 \Rightarrow \alpha_j \neq q_k$ for all $j \neq i$.⁴ If agent i with type θ_i receives an object with quality q and pays t for it, then his utility is given by $\theta_i q - t$.

Fix a (potentially random) allocation rule $\alpha : [0, 1]^n \times \Omega \rightarrow \mathcal{A}$ that depends on the agents' types $\theta_1, \dots, \theta_n$ and on randomness generated by the mechanism ω . For each i , let⁵

$$\varphi_i(\theta_i) = \mathbb{E}[\alpha_i(\theta_i, \theta_{-i}, \omega) \mid \theta_i]$$

denote the expected quality obtained by agent i , conditional on his type - this is also called the *interim allocation rule*. It is useful to also consider the quantile $s_i = F(\theta_i)$, and to define the *interim quantile allocation functions*

$$\psi_i(s_i) = \varphi_i(F^{-1}(s_i))$$

It is straightforward to show that an allocation α is part of a *Bayesian incentive compatible* (IC) mechanism if and only if each induced *interim quantile allocation* ψ_i is nondecreasing.

Denote by α^* the *assortative allocation* of agents to objects where the highest type gets highest quality, etc. and ties are broken by fair randomization. In our symmetric model, assortative matching α^* is incentive compatible, and induces the symmetric interim quantile allocation

$$\psi_i^*(s_i) = \psi^*(s_i) = \sum_{k=1}^n q_k \left[\frac{(n-1)!}{(k-1)!(n-k)!} (s_i)^{k-1} (1-s_i)^{n-k} \right].$$

⁴It is without loss of generality to assume that $m = n$ as we can always add objects with zero quality if $m < n$, or only sell the n highest-quality objects if $m > n$.

⁵A random allocation rule is a random variable defined on a probability space with outcomes (θ, ω) where ω describes the randomization in the mechanism and the probability measure prescribes that the agents types $\theta_1, \dots, \theta_n$ are i.i.d. distributed according to F and types θ are independent of the randomization of the mechanism ω .

A vector of interim allocations $\varphi = (\varphi_1, \dots, \varphi_n)$ where $\varphi_i : [0, 1] \rightarrow \mathbb{R}$, is *feasible* if there exists an allocation rule α that induces φ as its set of interim allocations, conditional on type. We restrict attention to symmetric interim allocation rules where $\varphi_i = \varphi$, $i = 1, 2, \dots, n$ and thus $\psi_i = \psi$, $i = 1, 2, \dots, n$.

In our terminology, Border's Theorem for the single object case, i.e., $q_n = 1$ and $q_k = 0$ for $k < n$, says that a symmetric and monotonic interim allocation φ is feasible if and only if the associated quantile interim allocation satisfies $\psi \prec_w s^{n-1}$.⁶ In this case, the assortative matching interim allocation $\varphi^*(\theta_i) = [F(\theta_i)]^{n-1}$ is the *efficient* allocation and hence $\psi^*(s_i) = (s_i)^{n-1}$.

Theorem 3 (Border's Theorem and BIC-DIC Equivalence). *In the ranked-items auction model the following holds:*

1. *A symmetric and monotonic interim allocation rule φ is feasible if and only if its associated quantile interim allocation $\psi(s) = \varphi(F^{-1}(s))$ is weakly majorized by the interim quantile allocation rule associated with the assortative allocation ψ^* .*
2. *For any symmetric, BIC mechanism there exists an equivalent, symmetric DIC mechanism that yields all agents the same interim utility, and that creates the same social surplus.*

Consider incentive-compatible mechanisms where the utility of the lowest type is zero (as required by individual rationality and revenue optimality). Denote by $J(\theta) = \theta - \frac{1-F(\theta)}{f(\theta)}$ the “virtual value” function. Then the expected revenue generated by a symmetric mechanism with interim allocation rule φ equals

$$n \int_0^1 J(\theta) \varphi(\theta) f(\theta) d\theta = n \int_0^1 J(F^{-1}(s)) \psi(s) ds.$$

Thus, by Theorem 3, the revenue maximization problem becomes

$$\max_{\psi \in \text{MPS}_w(\psi^*)} \int_0^1 J(F^{-1}(s)) \psi(s) ds$$

where ψ^* is the interim quantile allocation induced by assortative matching. The maximum is attained at an extreme point of $\text{MPS}_w(\psi^*)$, and by Corollary 1 there is $\hat{s} \in [0, 1]$ such that this extreme point is an extreme point of $\text{MPS}(\psi^* \cdot \mathbf{1}_{[\hat{s}, 1]})$ and equals zero on $[0, \hat{s}]$.

The above argument can be used in many other problems. One example is in Kleiner, Moldovanu, and Strack (2021), who briefly discuss an allocation problem with group-specific

⁶See also Maskin and Riley (1984) and Matthews (1984). This is not the original formulation. For connections to majorization see Hart and Reny (2015) (one-object).

quotas. Another example is Ashlagi, Monachou, and Nikzad, 2025, who consider dynamic settings where objects and agents arrive over time and have to be matched. They use the approach discussed above to design optimal queuing systems for the dynamic problem.

4.2 Contests

We now analyze a canonical contest design problem with a continuum of agents and prizes. Let F denote the distribution of types on $[0, 1]$, and let G denote the distribution of prizes awarded, also on $[0, 1]$. For simplicity, we assume that both F and G are strictly increasing, and consider allocation schemes where all prizes are distributed. If an agent with type θ obtains prize q and pays t for it, her utility is given by $\theta q - t$.

We analyze contests where each agent makes an effort (or submits a bid), and where agents are matched to prizes according to their bids. The assortative allocation is given by $\varphi^*(\theta) = G^{-1}(F(\theta))$, and is strictly increasing. It is implemented by the strictly increasing bidding equilibrium

$$t(\theta) = \theta \varphi^*(\theta) - \int_0^\theta \varphi^*(s) ds$$

The induced interim quantile allocation is given here by

$$\psi^*(s) = \varphi^*(F^{-1}(s)) = G^{-1}(F(F^{-1}(s))) = G^{-1}(s)$$

The agents' expected utility from the physical allocation of prizes is maximized by the assortative scheme,⁷ but agents need to waste resources (e.g., signaling costs, payments to a designer) in order to achieve it. Another feasible scheme is *random* matching where, independently of bids, everyone gets a prize equal to the expected value of the prize distribution μ_G . Expected utility from the physical allocation is smaller than under assortative matching, but random matching can be implemented without costs. The induced quantile distribution of prizes is given by

$$G_r(x) = \begin{cases} 0 & \text{if } x \leq \mu_G \\ 1 & \text{otherwise} \end{cases}$$

and thus $G_r \succ G \Leftrightarrow G_r^{-1} \prec G^{-1}$. Intermediate schemes can be obtained by *coarse* matching: for example, an agent with a bid in given quantile is randomly matched to a prize in the same quantile, i.e., he expects to obtain the average prize in that quantile. Coarse matching schemes balance output and bidding costs in less extreme ways than random or

⁷This follows from the rearrangement inequality of Hardy, Littlewood, and Pólya (1934). Under complete information, the set of feasible allocations is the set of measure-preserving mappings such that each subset of prizes is matched to a subset of agents of equal measure.

assortative matching, and may be superior for some objectives.

The Proposition below generalizes and complements several well-known, existing results in the contest and matching literature.⁸ These are obtained as immediate consequences of our theoretical insights together with the Fan-Lorenz Theorem.

Proposition 3.

1. *A matching scheme is feasible and incentive compatible if and only if the induced distribution of prizes G_{ic} satisfies $G_{ic}^{-1} \prec G^{-1}$.*
2. *Assume that the distribution of types F is convex. Then each type of the agent prefers random matching to any other scheme.*
3. *Random matching (assortative matching) maximizes the agents' expected utility if the distribution of types F has an Increasing (Decreasing) Failure Rate.*

4.3 Second-order stochastic dominance

We now relate majorization of cumulative distribution functions to second-order stochastic dominance and mean-preserving spreads.

Say that a cumulative distribution function F dominates a cumulative distribution function G in second-order stochastic dominance, $F \geq_{SOSD} G$ if $\int u(x) dF(x) \geq \int u(x) dG(x)$ for any concave continuous function u such that both integrals exist.

Proposition 4. *$F \geq_{SOSD} G$ if and only if $F \succ G$.*

Proof. Note that

$$\int_s^\infty [F(x) - G(x)] dx = -s[F(s) - G(s)] - \int_s^\infty x d[F - G](x) = \int_{-\infty}^\infty -\max\{s, x\} d[F - G](x)$$

Hence, if $F \geq_{SOSD} G$, the right-hand side is positive and F and G have the same expected value; therefore, $F \succ G$.

For the converse, suppose there is a concave continuous function u such that $\int u(x) dF(x) < \int u(x) dG(x)$. We can approximate u by a piecewise affine concave function $\bar{u}$ such that $\int \bar{u}(x) dF(x) < \int \bar{u}(x) dG(x)$. Moreover, there is an affine function a such that $\bar{u}$ is a convex combination of a and concave functions of the form $-\max\{s_i, x\}$ for $s_i \in \mathbb{R}$. But then $F \succ G$ implies $\int \bar{u}(x) dF(x) \geq \int \bar{u}(x) dG(x)$, a contradiction. Q.E.D.

⁸In recent work, Akbarpour, Dworczak, and Kominers (2024) discuss the main role played by the extreme points identified above for problems where a designer maximizes a weighted sum of revenue and social surplus given an arbitrary set of Pareto weights.

The next result goes back to Hardy, Littlewood, and Polya (generalized by Blackwell, Stein and Sherman, Cartier, and Strassen). It shows that if $F \succ G$ then G is a mean-preserving spread of F : there exists a kernel K such that, for every x , the expected value of $K(x, \cdot)$ is x and $\int K(x, \cdot) dF = G$. In particular, if X is a random variable with distribution F and Y is a random variable with distribution G then there exists mean-zero random variables Z such that $X = Y + Z$.

Proposition 5. *Let F and G have bounded support. Then $F \succ G$ if and only if G is a mean-preserving spread of F .*

Proof. Suppose G is a mean-preserving spread of F and let K denote a corresponding kernel. Jensen's inequality then implies for any concave function h ,

$$\int h(x) dG(x) = \int \int h(y) K(x, dy) dF(x) \leq \int h(x) dF(x)$$

and hence $F \succeq_{SSOD} G$. Proposition 4 implies $F \succ G$.

For the converse, suppose first that G is an extreme point of $\text{MPS}(F)$. By Theorem 1, there exists a collection of disjoint intervals $[\underline{x}_i, \bar{x}_i)$ indexed by $i \in I$ such that for a.e. $x \in [0, 1]$

$$G(x) = \begin{cases} F(x) & \text{if } x \notin \bigcup_{i \in I} [\underline{x}_i, \bar{x}_i) \\ \frac{\int_{\underline{x}_i}^{\bar{x}_i} F(s) ds}{\bar{x}_i - \underline{x}_i} & \text{if } x \in [\underline{x}_i, \bar{x}_i). \end{cases} \quad (6)$$

We can define a kernel as follows: if $x \in (\underline{x}_i, \bar{x}_i)$ for some $i \in I$, $K(x, \cdot)$ is the probability measure with support $\{\underline{x}_i, \bar{x}_i\}$ and expected value x , and if $x \notin \bigcup_i (\underline{x}_i, \bar{x}_i)$ then $K(x, \delta_x)$. It follows that G is a mean-preserving spread of F .

Now if G is a convex combination of extreme points of $\text{MPS}(F)$ then the corresponding convex combination of spreads does the job. For arbitrary $G \in \text{MPS}(F)$, let μ be a probability measure supported on the extreme points of $\text{MPS}(F)$ that represents G . Let ν be the induced probability measure on the set of kernels (see Lemma 1) and define $\bar{K}(x, A) := \int K(x, A) d\nu(K)$ for all $A \subseteq \mathbb{R}$ and x . Then $\int \bar{K}(x, \cdot) dF = \int H d\mu(H) = G$. Hence, G is a mean-preserving spread of F . Q.E.D.

4.4 Optimal stopping

The set of mean-preserving spreads of a given function also shows up in optimal stopping problems because this set equals the set of distributions of a stopped martingale. For an overview of this and related problems, see Obłój (2004).

In the following, let $(M_t : t \geq 0)$ be a continuous local martingale whose quadratic variation goes to infinity as t goes to infinity with $M_0 \sim F$. We consider randomized stopping times. To do so, let us assume there is a random variable uniformly distributed on $[0, 1]$ and independent of the martingale. A (randomized) stopping time is a function $T : \Omega \times [0, 1] \rightarrow \mathbb{R}$ if $\{(\omega, u) \in \Omega \times [0, 1] | T(\omega, u) \leq t\}$ is measurable with respect to the natural filtration times Borel.

Proposition 6. *Let G be a probability measure on $\mathbb{R}$ with bounded support. There exists a stopping time such that $M_T \sim G$ if and only if $G \in \text{MPS}(F)$.*

Proof. Let T be a stopping time such that $M_{T \wedge t}$ is a uniformly integrable martingale and let G denote its distribution. For any convex h , $(h(M_t))$ is a submartingale and Doob's optional stopping theorem implies $\int h dG = \mathbb{E}h(M_T) \geq \mathbb{E}h(M_0) = \int h dF$. Hence $G \in \text{MPS}(F)$.

For the converse, let G be an extreme point of $\text{MPS}(F)$. By Theorem 1, there exists a countable family $\{(\underline{x}_i, \bar{x}_i)\}_{i \in I}$ such that

$$G(x) = \begin{cases} F(x) & \text{if } x \notin \bigcup_{i \in I} [\underline{x}_i, \bar{x}_i) \\ \frac{\int_{\underline{x}_i}^{\bar{x}_i} F(s) ds}{\bar{x}_i - \underline{x}_i} & \text{if } x \in [\underline{x}_i, \bar{x}_i). \end{cases} \quad (7)$$

Let $A := \mathbb{R} \setminus [\bigcup_i (\underline{x}_i, \bar{x}_i)]$. Then the following stopping time T satisfies $M_T \sim \mu$: $T(\omega, u) = \inf\{t : M_t(\omega) \in A\}$, that is, stop as soon as the process enters A .

If G is a convex combination of extreme points, then the corresponding convex combination of stopping times induces G as the distribution of the stopped process. For arbitrary $G \in \text{MPS}(F)$ there exists a probability distribution μ supported on the extreme points of $\text{MPS}(F)$ that represents G . By Lemma 1, there is a corresponding distribution over stopping times, and we can define the corresponding randomized stopping time. (Alternatively, we can consider a sequence of convex combination of extreme points of $\text{MPS}(F)$ that converges to G . The corresponding sequence of stopping times converges along a subsequence (Baxter and Chacon, 1977), and the limits induces distribution G . Q.E.D.

4.5 Persuasion with Preferences over the Posterior Mean

We consider here the persuasion problem studied by Kolotilin (2018), Gentzkow and Kamenica (2016), and Dworzak and Martini (2019).

The state of the world ω is distributed according to a continuous distribution F on the interval $[0, 1]$, and a sender can reveal information about the state to an uninformed receiver. The sender chooses a signal (or Blackwell experiment) π that consists of a signal realization space S and a family of distributions $(\pi_\omega)_\omega$ over S , conditional on the state. By Bayes' rule

each signal induces a distribution of posteriors, and hence a distribution of posterior means. The receiver observes the choice of signal and the signal realization, and then chooses an optimal action that depends on the mean of the posterior, denoted here by x . The sender's indirect utility v is state independent and only depends on the posterior mean x .⁹ The following result (see Blackwell, 1953; Strassen, 1965; Kolotilin, 2018) characterizes which distributions of posterior means are feasible:

Proposition 7. *G is feasible if and only if $F \prec G$.*

Any signal is a “garbling” of the prior, and thus, for any signal π , the prior F is a mean-preserving spread of the generated distribution of posterior means G_π , i.e. $G_\pi \succ F$.

For the converse, suppose G_π is an extreme point of F . Then we can construct a signal that induces G_π : either you learn the state, there is pooling on an interval, or there is an interval on which two realizations are sent (by inverting the mean-preserving spread). If G_π is a convex combination of extreme points, we take the convex combination of the relevant signals; more generally, generalized convex combination (how to argue for measurability here?).

Hence, formally, the sender's problem is to choose a distribution over posterior mean beliefs of the receiver G that solves:

$$\max_{G \in \text{MPC}(F)} \int_0^1 v(x) dG(x).$$

As the objective is linear, a maximum is attained at one of the extreme points characterized in Theorem 2.¹⁰ This immediately implies that an optimal signal structure partitions the states in intervals such that, in each interval:

1. Either all states are perfectly revealed.
2. Or states are pooled, so that only one (deterministic) signal is sent for all states in this interval.
3. Or two different (potentially random) signals are sent for states in that interval, inducing two possible posterior means on this interval.

A signal structure is called *monotone partitional* if it partitions the state space into intervals such that each interval is either of type 1 or type 2; such an information structure either reveals the state perfectly, or sends the same signal for all states in an interval. While other information structures may be optimal, our result implies that an optimal

⁹This allows for the sender's payoff to depend on the action taken by the receiver.

¹⁰In the discussion paper version we also treat a Bayesian persuasion problem with an ex-ante informed agent, and we show how the insights of our Theorem 1 become relevant.

signal structure can still be implemented in a simple way by sending at most two signals on each interval. Arieli et al. (2023) independently obtained the same result - they call signal structures of type 3 *bi-pooled*.

4.6 Optimal Delegation

We now study a model of optimal delegation building on seminal work by Holmström (1984), Melumad and Shibano (1991), Alonso and Matouschek (2008), and Amador and Bagwell (2013). The state of the world θ is distributed according to a distribution F with support $[0, 1]$ and with density f . Its realization is privately observed by an agent. The action space is the real line. The principal can commit to an arbitrary mechanism.

The agent's utility from a deterministic action a in state θ is given by $u_A(\theta, a) = \theta a - 1/2a^2$, and the principal's utility is given by $u_P(\theta, a) = \gamma(\theta)a - 1/2a^2$, where $\gamma : [0, 1] \rightarrow \mathbb{R}$ is bounded. Both agent and principal have expected utility preferences.

A *direct mechanism* $m : [0, 1] \rightarrow \Delta(\mathbb{R})$ assigns to each agent's report a lottery over actions with finite mean and variance. A revelation principle applies and we focus on direct mechanism in which truthful reporting maximizes the agent's expected payoff. The agent's *indirect utility* from mechanism m is

$$U_m(\theta) := \max_{\theta'} \mathbb{E}_{M(\theta')} [\theta a - 1/2a^2].$$

This is a convex function that lies below the payoff the agent would get under full discretion, which is given by

$$h(\theta) := \max_a \theta a - 1/2a^2 = 1/2\theta^2.$$

One can show that any convex function U with $U \leq h$ is the indirect utility of some mechanism (Kleiner, 2025). To show this in the current context and establish the connection to majorization, we consider only mechanisms such that, for given $\theta_1, \theta_2 \in \mathbb{R}$, $U(\theta) = h(\theta)$ for all $\theta \leq \theta_1$ and $\theta \geq \theta_2$. Observe that for any mechanism we can find θ_1, θ_2 such that the principal's expected payoff is not changed if we impose that $U(\theta) = h(\theta)$ whenever $\theta \leq \theta_1$ or $\theta \geq \theta_2$. Using additional arguments, Kolotilin and Zapechelnnyuk (2019) and Kleiner, Moldovanu, and Strack (2021) determine values θ_1 and θ_2 such that we can impose this for any mechanism that might be optimal.

Proposition 8. *The principal's problem can be stated as choosing a function in $MPC(h')$ to maximize a linear objective function.*

Proof. Using that for any indirect utility under consideration $U(\theta_i) = h(\theta_i)$ for $i = 1, 2$ and $U(\theta) \leq h(\theta)$ for $\theta \in (\theta_1, \theta_2)$, we obtain $\int_{\theta_1}^{\theta} U'(z) dz \leq \int_{\theta_1}^{\theta} h'(z) dz = \int_{\theta_1}^{\theta} z dz$ for all $\theta \in [\theta_1, \theta_2]$

with equality for $\theta = \theta_2$ and therefore $U' \in \text{MPC}(h')$.

Conversely, for any extreme point g of $\text{MPC}(h')$ we can find a mechanism with indirect utility U that satisfies $U' = g$: Let g be given by (3) with the collection of intervals $\{[\underline{x}_i, \bar{x}_i] : i \in I\}$. If $\underline{y}_i = \bar{y}_i$ for all $i \in I$, this mechanism lets the agents choose among deterministic actions in $\mathbb{R} \setminus \bigcup_i (\underline{x}_i, \bar{x}_i)$; it is easy to check that this mechanism satisfying the requirement. If $\underline{y}_i < \bar{y}_i$ for some $i \in I$, for every such i the mechanism additionally offers the option of a lottery with expected value v_i and variance $(v_i - \underline{x}_i)(\underline{y}_i - \frac{1}{2}(\underline{x}_i + v_i))$. This variance is nonnegative since $g \in \text{MPC}(h')$ and ensures that type $\underline{y}_i$ is indifferent between deterministic action $\underline{x}_i$ and this lottery and likewise type $\bar{y}_i$ is indifferent between deterministic action $\bar{x}_i$ and this lottery. Then mix over those mechanisms to obtain arbitrary U' in this set.

If g is a convex combination of extreme points, a convex combination of the corresponding mechanisms induces an interim utility satisfying $U' = g$. And for arbitrary $g \in \text{MPC}(h')$, there is a probability distribution μ supported on the extreme points of MPC that represents g . Lemma 1 implies that there is a corresponding probability distribution over mechanisms and therefore a mechanism with indirect utility satisfying $U' = g$.

Finally, note that the principal's payoff is linear in U' : $U(\theta) = \mathbb{E}_{m(\theta)}[\theta a - 1/2a^2]$ and $U'(\theta) = \mathbb{E}_{m(\theta)[a]}$. Therefore, the principal's payoff given type θ and a mechanism with indirect utility U is

$$\mathbb{E}_{m(\theta)}[g(\theta)a - 1/2a^2] = g(\theta)U'(\theta) + U(\theta) - U'(\theta)\theta = g(\theta)U'(\theta) + h(\theta_1) + \int_{\theta_1}^{\theta} U(s) ds - U'(\theta)\theta,$$

where the last equality uses $U(\theta_1) = h(\theta_1)$.

Q.E.D.

We mention three implications of this result: First, it yields qualitative insights into optimal mechanisms. Observe that a solution can always be found at an extreme point of $\text{MPC}(h')$ and recall the characterization of extreme points in Theorem 2. If in every interval $\underline{y}_i = \bar{y}_i$ then the mechanism is deterministic and can be implemented by letting the agent choose her favorite action in the set $\mathbb{R} \setminus \bigcup_i (\underline{x}_i, \bar{x}_i)$. Generally, whenever $\underline{y}_i < \bar{y}_i$, an additional stochastic action has to be added. Optimal delegation mechanisms therefore sometimes involve deliberate randomization by the principal (see Kovác and Mylovanov (2009) and Alonso and Matouschek (2008) for earlier examples). But our result above significantly reduces the class of uniquely optimal stochastic mechanisms: any extreme (and thus exposed) point will use at most one nondegenerate lottery on each of the intervals $(\underline{\theta}_i, \bar{\theta}_i)$, and any stochastic extreme point will have a discontinuous mean-action function.

Second, this result implies that we can use Proposition 2 to characterize when a particular extreme point is optimal.

Third, this result implies that there is a close connection between certain delegation and

persuasion problems in that they share the same mathematical structure. Kolotilin and Zapechelnnyuk (2025) show that the two problems are in fact equivalent, even with more general preferences than the ones considered here, as long as types and actions are one-dimensional.

In contrast, in the multidimensional case, neither the delegation problem nor the persuasion problem can be formulated reasonably using a majorization constraint, and the two problems are no longer equivalent. Kleiner (2025) analyzes multidimensional delegation and shows that the design problem can still be formulated as choosing a convex function below a given function.

5 Recent related extreme point characterizations with economic applications

Following the general methodology of extreme point characterizations, recent research has extended these results to address additional constraints, multidimensional settings, and alternative orders relevant to various economic applications.

5.1 Additional Constraints

In some economic applications, in addition to the monotonicity and majorization constraints, additional constraints are relevant. Nikzad (2022) analyzes the extreme points of the set of functions that are either in $\text{MPS}(f)$ or in $\text{MPC}(f)$ and satisfy finitely many additional linear constraints (see also Candogan and Strack (2023)). These extreme points are closely related to the extreme points characterized above, but can have up to one additional jump discontinuity per extra constraint. Nikzad (2022) then applies his results to auction problems in which the designer has to satisfy additional constraints (e.g., the welfare generated by the auction has to lie above some bound), to a redistribution problem based on Dworzak, Kominers, and Akbarpour (2021), and to a procurement problem based on Gershkov, Moldovanu, et al. (2021).

Taking a different but related approach, Augias and Uhe (2025) consider the set of convex functions that lie between two given functions and characterize its extreme points. To connect this to the characterization above, take the given convex functions to be the integrals of nondecreasing functions f and g , where $f \succ g$. Their result then characterizes the extreme points of $\text{MPS}(f) \cap \text{MPC}(g)$. They apply their result to delegation and persuasion problems with outside options.

5.2 Multidimensional Extensions

While majorization elegantly captures various economic design problems in onedimensional spaces, the resulting notions in multidimensional spaces often differ. One example is the pair of delegation and persuasion problems that give rise to the same constraint set MPC. To illustrate the differences between the corresponding problems in multidimensional settings, observe that our characterization above implies that the extreme points of the set of convex functions that lie below a given convex function have a tractable structure. In contrast, in multidimensional spaces, the set of extremal convex functions is extremely large and, in fact, dense in the set of convex functions (Johansen, 1974; Bronshtein, 1978; Lahr and Niemeyer, 2024). This implies that a mere characterization of the extreme points loses many of the advantages we used above and shows that different analytical tools are required for multidimensional problems of this form (e.g., results on when particular solutions are optimal instead of results on which solutions can be optimal).

On the other hand, our characterization of the extreme points of the set of probability distributions that are mean-preserving contractions of a given probability distribution can be extended to multidimensional spaces. Moreover, these extreme points retain their relatively simple structure that we observed in one dimension: every extreme point with finite support is characterized by a partition of the state space into convex sets such that the original distribution is contracted, on each partition element, to a probability distribution with affinely independent support (Kleiner, Moldovanu, Strack, and Whitmeyer, 2024). This illustrates a fundamental difference between persuasion and delegation problems in multidimensional spaces.

5.3 Alternative Orderings

Moving beyond majorization orderings, recent work has explored alternative ways to order functions that arise naturally in economic applications. K. H. Yang and Zentefis (2024) characterize the extreme points of the set of nondecreasing functions that lie pointwise between two given nondecreasing functions. They apply their results to gerrymandering, quantile-based persuasion (see also Kolotilin and Wolitzky, 2024), overconfidence, and security design.

F. Yang and K. H. Yang (2025) analyze nondecreasing functions $f : [0, 1]^n \rightarrow [0, 1]$ and their extreme points. They also analyze tuples of nondecreasing functions $(q_1, \dots, q_n)$, where $q_i : [0, 1] \rightarrow [0, 1]$, such that there is a function $f : [0, 1]^n \rightarrow [0, 1]$ whose i th marginal is q_i (that is, $\int f(x_i, x_{-i}) dx_{-i} = q_i(x_i)$) and characterize the extreme points of this set. They apply their results to analyze interim efficient mechanisms in bilateral trade, asymmetric reduced form auctions, public good problems, and information design with privacy constraints.

These extensions demonstrate both the versatility of the extreme point approach and the rich variety of constraint structures arising in modern economic design problems.

Appendix

Lemma 1. *Let X, Y be completely metrizable topological vector spaces and let $T : X \rightarrow Y$ be affine, continuous, and onto. Then the inverse of T has a measurable selection. Moreover, for any $\mu \in \Delta(Y)$ there exists $\nu \in \Delta(X)$ such that $\int g \circ T d\nu = \int g d\mu$ for all g such that either integral exists.*

Proof. T is an open mapping by Thm. 5.18 in Aliprantis and Border (2006). Therefore, its inverse, denoted by S , is lower hemicontinuous (Thm. 17.7 in Aliprantis and Border, 2006). It follows that S is weakly measurable (Def. 18.1 in Aliprantis and Border, 2006). Since T is continuous and onto, S is closed-valued and nonempty-valued. By Thm. 18.13 in Aliprantis and Border (2006), it has a measurable selection, denoted by s . We define the pushforward of μ under s by $\nu(E) := \mu(s^{-1}(E))$ for any Borel $E \subseteq X$. The claim follows from the change of variable formula for the pushforward measure. Q.E.D.

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